

Structured Scan-to-Model Discrepancy in Spacecraft LiDAR Registration: Local Pose Effects and Historical Compensation

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ABSTRACT

Systematic differences between LiDAR scans and a nominal spacecraft model can displace registration estimates from the reference pose. We examine this local inconsistency and test compensation using calibrated observations of the same target. A least-squares projection with fixed correspondences identifies a pose-active residual component. Across 1456 EPOS-Lid frames, the median ratio of pose-active to total root-mean-square (RMS) residual magnitude ranges from 0.30 to 0.39 across four analyzed subsets. The first-order translation prediction has median directional cosine similarities of 0.77–0.95 with the displacement obtained after ICP, compared with 0.959–0.995 for a finite-difference predictor with rematching. Controlled experiments with exact reference poses show that comparable residual RMS values can accompany markedly different translation errors. A 24-patch discrepancy field calibrated on trajectory VI reduces median reference-relative translation error relative to uncorrected ICP on all four main analysis subsets. The support rule retains 6.4%, 34.2%, and 28.5% of trajectories IV, II, and III, respectively; the median error on retained III frames decreases from 75.0 to 41.3 mm. Leave-one-block-out calibration on VI retains approximately 99% of the in-sample paired translation benefit, which also remains positive across the tested parameter and initialization settings. However, the additional benefit over a global historical translation correction depends on the trajectory. Adding Patch to Huber reduces translation error but produces median framewise increases in rotation error of 0.052°–0.720°. These results support local translation-error reduction for the calibrated target within the evaluated support, with limited coverage and trajectory-dependent transfer and rotation effects.

1. INTRODUCTION

Close-proximity operations around non-cooperative spacecraft require reliable relative pose estimates for rendezvous, inspection, servicing, and debris-removal tasks [1]. At close range, LiDAR provides metric three-dimensional measurements that can be registered to a known target model using ICP-type estimators [2, 3, 7–9]. Existing spacecraft-registration studies mainly address initialization, partial or sparse observations, correspondence quality, ambiguity, and tracking [5, 6]. A local consistency problem arises when the measured scan differs systematically from the nominal target representation: optimization initialized at the reference pose can move the estimate toward a different rigid transform.

We do not attribute the residuals observed in real spacecraft LiDAR data to a single physical source. It can contain target/model discrepancy, scan-integration effects, reference-pose and extrinsic uncertainty, sampling differences, and surface-response effects. We therefore use the term *structured scan-to-model discrepancy*. The central question is not whether reference-model uncertainty exists, but which part of the residual can act through the local rigid-motion degrees of freedom and whether repeated observations of the same target can be used to reduce that local effect.

Our geometry-level compensation differs from post-hoc correction of an estimated pose. The translation-state baselines considered here predict a translation-error vector and subtract it from the translation component of the registration estimate. Our primary Patch method instead estimates a target-frame spatial discrepancy field from calibrated observations of the same spacecraft, modifies the nominal geometry, and reruns correspondence search and optimization. Under specific assumptions, a spatially constant geometric correction can be equivalent to a translation-state correction. A spatially varying field can also affect rotation and correspondence selection. The spacecraft setting considered here involves substantial changes in viewpoint, partial visibility, repeated use of a nominal target model, and calibration data for the same target.

This study addresses three questions. First, how much of the reference-pose residual lies in the local rigid-motion subspace, and how closely does the associated first-order direction agree with the displacement realized after rematching? Second, can a fixed same-target spatial field reduce reference-relative translation error on trajectories not used to fit the field? Third, under which support, view-conditioning, robust-loss, and initialization conditions do these corrections lose their translation benefit or increase rotation error?

The contributions are threefold. (1) We compare, on the same real spacecraft LiDAR frames, a fixed-correspondence pose-active projection, a rematching finite-difference predictor, and the realized local displacement. (2) We use

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controlled exact-reference experiments to show that similar residual RMS can accompany different translation errors and that robust losses respond differently to coherent and contamination-like discrepancy. (3) We evaluate same-target historical correction at global, spatial, view-conditioned, and pose-state levels, jointly reporting external support coverage, cross-trajectory transfer, VI held-out calibration, coarse-initialization robustness, computational cost, and translation–rotation trade-offs. The conclusions are limited to the evaluated data support, initializations, solver settings, and hardware configuration.

2. RELATED WORK

2.1. Spacecraft LiDAR registration

Relative pose estimation for non-cooperative spacecraft has been studied using monocular and stereo cameras, as well as active depth sensors such as time-of-flight cameras and LiDAR [1]. For known targets at close range, LiDAR permits direct registration of metric 3D measurements to a target model. Woods and Christian [2] combined a global 3D descriptor with ICP refinement; Liu et al. [3] used flash-LiDAR point clouds and a satellite model for pose acquisition and ICP tracking; He et al. [5] embedded point-cloud features in a particle-filter tracker. More recently, Tecchia et al. [6] compared correspondence-based global registration methods for known non-cooperative spacecraft and refined their estimates with point-to-plane ICP.

These studies primarily target acquisition, correspondence quality, partial visibility, ambiguity, or tracking. The present work instead starts from a controlled local regime and asks whether a nominal model that is not fully consistent with the measured scan can itself induce reference-relative displacement. Reference-pose initialization is therefore an experimental probe of objective consistency, not a proposed deployment front end.

2.2. Robust registration and local ICP sensitivity

Rigid point-cloud registration ranges from classical point-to-point ICP [7] and point-to-plane ICP [8] to generalized ICP [9]. Robust variants address partial overlap, incorrect correspondences, and large residuals associated with outliers. Trimmed ICP minimizes a least-trimmed-squares objective over a retained subset of correspondences [13]; robust M-estimators reduce the influence of large residuals, with behavior depending on the residual regime and tuning [14]; TEASER uses truncated least squares and certifiable estimation for high outlier rates [15]. Broader surveys emphasize that sampling, matching, rejection, weighting, and optimization are coupled design choices [16].

Local Hessian and Jacobian structure has also been used to analyze ICP uncertainty [17]. Bonnabel et al. [18] showed that ICP covariance estimates based on fixed-correspondence linear least-squares models can be invalid when rematching is ignored, particularly for point-to-point ICP. We use the fixed-correspondence projection only as a

local diagnostic and explicitly compare it with an empirical finite-difference predictor of the rematching objective. We evaluate robust estimators as baselines for coherent structured discrepancy.

2.3. Scan-to-CAD discrepancy and reference-model uncertainty

Manufacturing metrology and scan-to-CAD registration provide closely related work on model discrepancy. Scan-to-CAD methods use geometric features and constraints to stabilize alignment when measurements are incomplete or initialization is uncertain [11]. Decker et al. [10] showed that unconstrained ICP can absorb part of the actual shape deviation of manufactured parts, so the estimated rigid transform can mask geometric discrepancy that inspection aims to measure. This is conceptually important here: a structured scan-to-model discrepancy need not remain only in the residual; part of it can appear as an estimated pose change.

Reference-model uncertainty can also be represented directly inside LiDAR registration. Xu et al. [12] modeled uncertainty in generalized LoD2 city models during outdoor LiDAR registration. The present study does not claim that model uncertainty or geometric correction is new in itself. Its narrower contribution is to examine the local rigid-motion component on spacecraft LiDAR data, test whether comparable residual magnitudes can be associated with different local pose effects, and evaluate whether a discrepancy field learned from repeated observations of the same non-cooperative target transfers across trajectories.

2.4. Historical state correction versus geometry-level compensation

Systematic error may be corrected after state estimation or in the geometry that defines the objective. Vlaeyen et al. [19], for example, modeled systematic laser-line-scanner error and compensated measured geometry before downstream dimensional evaluation. In spacecraft registration, a historical state-space method can predict translation bias from viewing conditions and subtract that bias from a new pose estimate. Our Global-Vector, DBS, and Vector-DBS baselines represent this route. Patch and Full instead alter nominal geometry before rematching.

These two routes coincide only in special cases. A target-wide constant geometric translation can resemble a translation-state correction, but a spatially varying field changes correspondence geometry and couples translation and rotation. Full additionally conditions the correction field on the query viewing state using nearby historical observations. The scientific comparison in this paper is therefore across different representations and applications of historical information: global and view-conditioned translation-state corrections, and spatial geometry corrections with or without view conditioning.

Table 1

Positioning relative to representative prior work. The present contribution is the combination of local residual-to-pose analysis, same-target historical geometry correction, and evaluation of cross-trajectory transfer and translation–rotation trade-offs in spacecraft LiDAR registration.

Work	Primary setting	Reference-model discrepancy	Residual-to-pose analysis	Historical geometry correction / transfer
Censi; Bonnabel et al. [17, 18]	Generic ICP uncertainty	Not primary focus	Local Hessian/Jacobian analysis with rematching caveats	No target-specific historical correction
Decker et al. [10]	Manufacturing scan-to-CAD inspection	Shape deviation can be absorbed by rigid alignment	Evaluates registration-induced masking of geometric deviation	Manufacturing setting; no spacecraft trajectory transfer
Xu et al. [12]	Outdoor LiDAR to semantic city models	Explicit uncertain reference geometry	Model uncertainty enters registration constraints	No repeated-target historical correction
Present study	Same-target spacecraft LiDAR	Structured scan-to-nominal-model discrepancy	Pose-active projection, rematching finite differences, matched-RMS tests	Global/spatial/view-conditioned history with cross-trajectory support and 6-DoF trade-offs

3. METHODOLOGY

3.1. Local scan-to-model discrepancy and pose-active component

Let T_{ref} map the target frame to the sensor frame. Each measured point is first expressed in the target frame as

$$q_i = T_{\text{ref}}^{-1} p_i, \quad \hat{T} = T_{\text{ref}} C^{-1}, \quad (1)$$

where p_i is a measured point in the sensor frame, q_i is its reference-transformed target-frame coordinate, and C is the rigid correction applied to that reference-transformed scan; $C = I$ corresponds to the dataset reference pose. Controlled experiments use an exact reference pose; real-data “error” is therefore reference-relative, not absolute ground-truth error. The local chart is $\xi = [t^T, \omega^T]^T$, with translation in metres and Rodrigues rotation in radians, and

$$q_i(\xi) = R(\omega)q_i + t. \quad (2)$$

The primary registration objective is the mean squared point-to-point distance to the nearest nominal model points,

$$E(C) = \frac{1}{N_t} \sum_i \|Cq_i - m_{c(i,C)}\|_2^2, \quad (3)$$

which is minimized using ICP, with correspondences $c(i, C)$ recomputed at each iteration. To isolate the local effect of model discrepancy, correspondences are frozen only for the analytic diagnostic. With residual $\delta = r(0)$, Jacobian J , and positive-semidefinite weight matrix W , define the fixed-correspondence weighted quadratic model

$$r(\Delta\xi) \approx \delta + J\Delta\xi, \quad \phi(\Delta\xi) = \frac{1}{2}(\delta + J\Delta\xi)^T W (\delta + J\Delta\xi), \quad (4)$$

The weighted projection onto the residual subspace generated by local rigid motion is

$$P_{J,W} = J(J^T W J)^{\dagger} J^T W, \quad \delta_{\text{PA}} = P_{J,W} \delta, \quad (5)$$

and the minimum-norm solution in the stated pose coordinates is

$$\Delta\xi_{\text{FO}} \approx -H_{\text{GN}}^{\dagger} g_{\text{ref}}. \quad (6)$$

We call δ_{PA} the *pose-active* component. This term is descriptive: Eq. (6) is a fixed-correspondence local approximation, not a global ICP model. Translation and rotation blocks are interpreted separately because they have different units. The empirical predictor used later is computed independently from central finite differences of the registration objective, with correspondences recomputed at every perturbed pose; it is not identified with g_{ref} .

3.2. Historical spatial correction and its local link

We partition the nominal model once into 24 patches by applying MiniBatchKMeans to the concatenated position and weighted-normal features $[x, y, z, 0.3n_x, 0.3n_y, 0.3n_z]$, with the random seed fixed at 42. The partition is a low-dimensional spatial index, not semantic segmentation. For calibrated scan s , target-frame discrepancy vectors are

$$d_{i,s} = q_{i,s} - m_{c(i,s)}, \quad (7)$$

and the patch correction is the mean discrepancy over all historical points assigned to that patch:

$$\mu_j = \frac{\sum_s \sum_{i: c(i,s)=j} d_{i,s}}{\sum_s N_{s,j}}, \quad j = 0, \dots, 23. \quad (8)$$

Here $N_{s,j}$ denotes the number of points from scan s assigned to patch j . All 501 trajectory-VI development scans are used

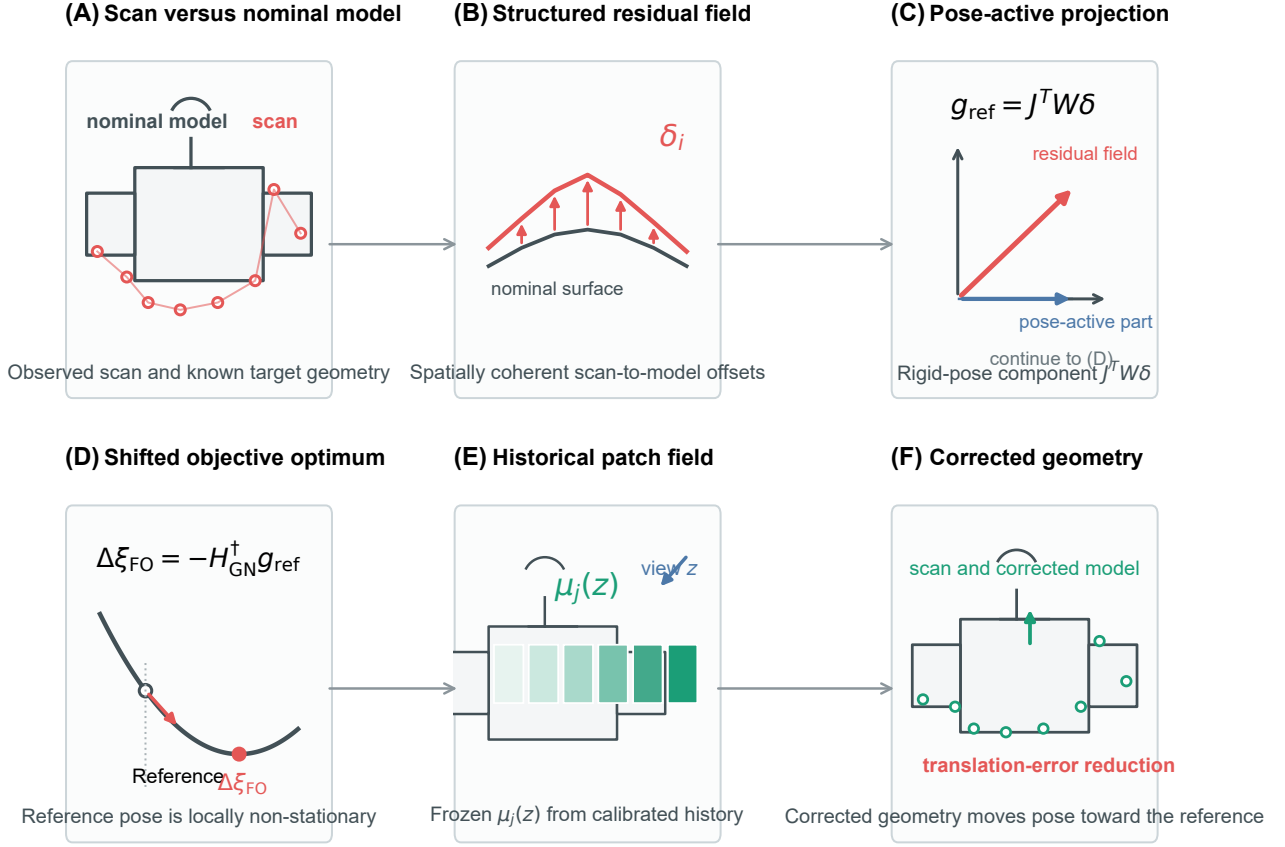


Figure 1: Mechanism and compensation. (A,B) A scan can contain spatially organized discrepancy from nominal geometry. (C) The residual-space component is $P_{J,W} \delta$ and the local weighted quadratic model uses $g_{\text{ref}} = J^T W \delta$ and $H_{\text{GN}} = J^T W J$. (D) The fixed-correspondence model predicts a local first-order pose step, distinct from the empirical rematching diagnostics. (E) Patch uses a fixed historical spatial field; Full optionally uses a view-conditioned field. (F) Corrected geometry is used for re-registration, and translation and rotation errors are evaluated relative to the reference pose.

to fit this field; empty scan-patch cells are ignored. The Global baseline applies one constant discrepancy vector to the entire nominal model. Patch modifies the model as

$$m_i^{\text{corr}} = m_i + \mu_{\pi(m_i)}, \quad (9)$$

and registration is then rerun against the corrected geometry. Model point order is unchanged, normals are not re-computed, and no smoothing or boundary interpolation is applied.

The local relation between Patch and the pose-active analysis is explicit. If $A_{\pi} \mu$ stacks the correction vectors assigned to the current correspondences, then under fixed correspondences and fixed J, W ,

$$\Delta \xi_{\text{corr}} \approx -H_{\text{GN}}^+ J^T W (\delta - A_{\pi} \mu). \quad (10)$$

Under this fixed local model, Patch changes the predicted pose step through $J^T W A_{\pi} \mu$. It can reduce the predicted step when this contribution compensates for the corresponding component of $J^T W \delta$. This approximation may fail after changes in correspondences or patch assignments, changes in J , or corrections large enough for higher-order terms to matter. In formulations that use surface normals, inconsistent normals introduce an additional source of error. In

particular, Eq. (10) does not guarantee lower translation error, lower rotation error, or convergence after finite-budget ICP. These failure modes motivate the rematching, transfer, and rotation analyses below.

The primary configuration—24 patches, MiniBatchK-Means on $[x, y, z, 0.3n_x, 0.3n_y, 0.3n_z]$, and clustering seed 42—is held fixed throughout the primary analysis. A fixed-grid sensitivity analysis varies the patch count $K \in \{12, 18, 24, 32, 48\}$, normal-feature weight $w \in \{0, 0.15, 0.30, 0.60, 1.00\}$, and clustering seed in $\{0, 1, 2, 42, 20240910\}$. For each alternative setting, the partition and VI calibration field are rebuilt while the solver, support rule, frame roster, and metrics remain unchanged. These sweeps test stability around the frozen implementation; they are not used to redefine the main configuration or to claim optimality.

3.3. View-conditioned extension and comparator hierarchy

The optional Full extension conditions the patch field on range and target-frame viewing direction, $z = (r, u)$. Full uses a kernel over the $k = 16$ nearest historical views from VI. The value of k was selected from $\{8, 16, 32, 64\}$

using VI before external evaluation. Full differs from Patch in two ways: it introduces view-neighbour weighting and uses equal-scan rather than point-count aggregation. It is therefore treated as a compound extension rather than a pure view-conditioning ablation. The support threshold $\tau_{\text{support}} = 0.44033$ is the 95th percentile of the nearest-training-view distances obtained by leaving out each VI block in turn. It defines the external subsets included in the main analysis.

Oracle Full queries the field with the dataset reference view. Estimated Full first registers the scan to the nominal model. Under the transform convention in Eq. (1), if the first pass returns local correction C_1 , the estimated pose is $\hat{T}_1 = T_{\text{ref}} C_1^{-1}$; this pose supplies the view query and the initialization for the second registration. Because that first pass is itself initialized at the reference pose, Estimated Full quantifies only the local cost of replacing the reference-view query with an estimated-view query and warm start. It does not test coarse acquisition or non-reference initialization. This experiment evaluates local view-query substitution. The separate initialization study tests fixed perturbations up to 300 mm and 15° (Sec. 4), but neither experiment establishes global acquisition performance.

The historical comparators differ in where correction is applied. Global, Patch, and Full alter model geometry before registration. DBS predicts a translation direction and magnitude from the same VI view neighbourhood and subtracts it in state space; Vector-DBS directly averages historical translation-bias vectors; Global-Vector subtracts one target-wide VI mean translation-bias vector. All three state-space baselines leave rotation unchanged. These baselines test whether spatial geometry provides a translation benefit beyond that obtained from historical translation-state correction.

4. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

4.1. Dataset, support, and information use

The real-data study uses the public EPOS-Lid benchmark [4] and its distributed target point cloud (67,870 points). The EPOS pose associated with each accumulated scan is used as the dataset reference. Verified per-point timestamps are unavailable, so intra-scan motion cannot be reconstructed; real residuals remain composite quantities that can include geometry, scan integration, calibration/reference uncertainty, sampling, and surface response.

Only VI is used for field fitting and parameter selection. IV and II are external cross-trajectory transfer sets. Trajectory III was reserved for evaluation of the fixed configuration; subsequent analyses of its retained subset are treated as secondary. V is used only for out-of-support diagnostics. The same support mask is applied to every compared method. Consequently, accuracy and coverage must be reported together rather than interpreting retained-frame accuracy as full-trajectory performance.

The external retention rates are low: 6.4% on IV, 34.2% on II, and 28.5% on III. This is a central limitation, especially for IV, and all main accuracy claims refer only to these retained subsets.

4.2. Registration, controlled tests, and sensitivity diagnostics

All main accuracy results use point-to-point ICP solved by weighted Kabsch/SVD with one-way nearest-model correspondences, a default 40-iteration cap, and the frozen 0.30 m/ 15° path safeguard. Huber uses $\delta_H = 1.345$ with a per-iteration MAD scale and a floor equal to the median nominal nearest-neighbour spacing (8.106 mm); Trim retains the 80% smallest residuals. Patch+Huber and Patch+Trim combine the frozen Patch geometry with the unchanged robust settings. No trajectory-specific retuning is allowed.

The predictor uses central finite differences with perturbation magnitudes of 5 mm for translation and 0.25° for rotation, recomputing correspondences at each perturbed pose. Reference-pose-started coordinate-axis probes test whether finite perturbations decrease the rematching objective. A direct fixed-correspondence evaluation on 1456 analyzed VI/IV/II/III frames computes $P_{J,W}\delta$ and Eq. (6) on the same frames used to measure realized local displacement.

The controlled exact-reference study uses three asymmetric spacecraft-like geometries and five structured mismatch families: coherent appendage translation (D1), missing nominal structure (D2), local bus offset (D3), one-sided span growth (D4), and coherent appendage tilt (D5), plus isotropic noise. Each geometry-mismatch-dose condition has 20 repetitions. A matched-severity analysis pairs cross-family conditions whose measured residual RMS differs by at most 5%; it is a conditional counterexample to RMS sufficiency, not a causal isolation of residual organization.

Several sensitivity diagnostics are reported without changing the frozen main configuration. First, 40 fixed frames per trajectory are rerun with iteration caps 40, 80, and 160. Second, the frozen 20-frame-per-trajectory initialization roster is evaluated with the same sign-balanced deterministic perturbation family from 10 mm/ 0.5° through 300 mm/ 15° ; the fraction of runs with final errors satisfying $e_t \leq 50$ mm and $e_R \leq 2^\circ$ is reported as a descriptive threshold-based performance measure, not a mission tolerance. Third, the Patch-count, normal-feature-weight, and clustering-seed grids defined in Sec. 3.2 are replayed on the unchanged VI/IV/II/III frame roster; III is retrospective sensitivity evidence only and is never used for parameter selection. Finally, two implementation ablations address the Patch+Huber rotation trade-off: model normals are re-estimated on the corrected geometry with the same local-PCA $k = 16$ protocol used for the nominal model, and a single neighbourhood-averaging step with k fixed at 16 is applied to the patch correction field as a boundary-smoothing diagnostic. These probes test robustness of the frozen local pipeline; no global initializer is introduced.

A six-fold VI leave-one-block-out (LOBO) calibration check holds out each frozen contiguous VI block in turn.

Table 2

EPOS-Lid data roles and fixed analysis support. Range statistics are min–median–max over the analyzed subset. The same nearest-view support mask is used for all compared methods in the main VI/IV/II/III comparisons, so accuracy claims apply to the retained subset rather than the full trajectory. For V, coverage denotes the fraction analyzed for exploratory diagnostics, not the fraction satisfying the main support rule.

Set	Avail.	Analyzed	Coverage (%)	Blocks	Range (m)	Analysis role	Information use
VI	501	501	100.0	6	8.747–12.095–14.802	Development/calibration	Field construction, k selection, method development
IV	2428	156	6.4	4	14.773–14.817–15.707	External transfer	No field fitting or parameter tuning; inspected before reserved III
II	1253	428	34.2	9	8.545–10.731–15.691	Difficult external transfer	No field fitting or parameter tuning; inspected before reserved III
V	1868	1868	100.0	38	2.789–4.420–7.756	Out-of-support exploratory	Support-boundary analysis only
III	1302	371	28.5	8	8.250–11.509–15.464	Reserved-stage support subset	One-time frozen analysis; secondary block statistics

The 24-patch partition remains fixed, while the per-patch discrepancy mean is rebuilt from the other five blocks and each of the 501 VI scans is scored exactly once out of fold; train/test scan identifiers are asserted disjoint for every fold. This evaluates out-of-fold field calibration while holding the nominal-model partition fixed. A separate computational benchmark uses the same frozen 20-frame-per-trajectory roster (80 frames total) on one CPU workstation with single-thread BLAS and nearest-neighbour queries, one warm-up and five timed calls per frame/method, and fresh-process peak-RSS measurements. For Raw and Patch, static model geometry and the corresponding KD-tree are built once and cached outside the timed per-frame region. Full includes its per-frame view query, corrected-model construction, and KD-tree construction in the timed latency; Estimated Full additionally includes both the nominal and corrected registrations. One-time Patch clustering, field assembly, corrected-model construction, and KD-tree construction are reported separately. This accounting separates reusable setup from per-frame deployment latency and reports hardware-specific cost rather than real-time capability.

A timing diagnostic uses only inter-frame pose increments and timestamps. Weak association with inter-frame angular speed cannot rule out intra-scan motion because point-level timing is missing. Low-dimensional VI-fit nuisance corrections are retained as secondary attribution checks, but they do not identify the physical source of the EPOS residual.

4.3. Metrics and statistics

Reference-relative translation and rotation errors are reported as

$$e_t = \|\xi_{1:3}\|_2 \times 1000 \text{ mm}, \quad e_R = \|\xi_{4:6}\|_2 \times 180/\pi \text{ deg.} \quad (11)$$

VI uses six contiguous view-travel blocks; retained IV/II/III frames are grouped by temporal rank into 4/9/8 blocks under the primary 50-frame convention. We compute percentile bootstrap intervals for each method's median error by resampling complete blocks with replacement (2000 resamples). Paired effects are computed within frame or block as stated for each comparison. Exact sign tests are secondary and

unadjusted. Because the block counts remain small, inference is conditional on the analyzed support and should not be read as a population-level guarantee. As a sensitivity check, the external trajectories are re-aggregated using 25-, 50-, and 100-frame blocks on the identical frame roster; no registration is rerun and no block length is selected by significance.

5. RESULTS

5.1. Structured residuals produce a measurable local pose response

The VI residual field is spatially nonuniform (Fig. 2), but residual magnitude alone is not a pose metric. In the reference-pose finite-descent test, 1445/1503 translation-axis probes and 1348/1503 rotation-axis probes decrease the rematching objective at 10 mm and 0.5°, respectively. The matched self-null finite-step response remains orders of magnitude smaller than the real-data response. These probes show local inconsistency of the nominal objective at the tested scale; they do not prove a distinct global optimum or global convergence.

The same-sample fixed-correspondence analysis provides a more direct local mechanism check. Across VI/IV/II/III, the median $\text{RMS}_{\text{PA}}/\text{RMS}$ ratios are 0.359, 0.392, 0.299, and 0.346. Median cosine similarity between the analytic first-order translation direction and realized displacement is 0.942, 0.946, 0.771, and 0.809; the rematching finite-difference predictor gives 0.959, 0.973, 0.990, and 0.995. The analytic step underestimates the translation displacement obtained after the prescribed ICP iteration budget; the median ratio of predicted to realized magnitude ranges from 0.25 to 0.42 across the four subsets. The first-order step provides a local approximation to the translation direction, while the finite-difference predictor with rematching agrees more closely with the displacement obtained after ICP.

The nuisance and timing diagnostics do not justify a unique physical interpretation of the real residual. VI-fit low-dimensional corrections can reduce the translation-driving gradient more than they reduce residual RMS on IV/II, but they fail to transfer uniformly to V. Inter-frame angular speed has only weak within-trajectory rank association with Raw

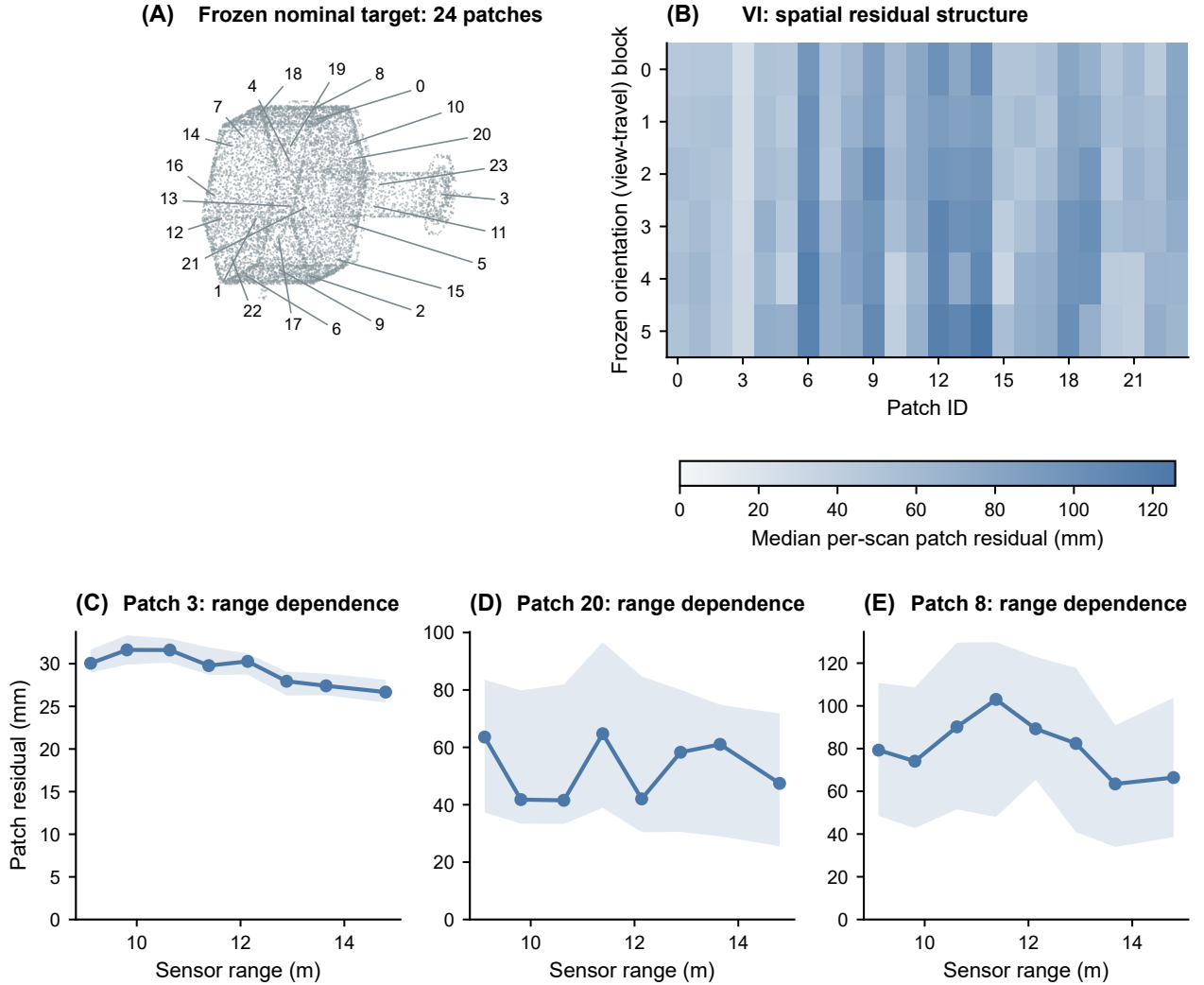


Figure 2: Structured real scan-to-model discrepancy. (A) Fixed nominal target partitioned into 24 zero-based patches. (B) For each patch and VI block, the color shows the median across scans of the within-patch median residual magnitude. (C–E) Range dependence for representative patches 3, 20, and 8. Each observation is one supported VI scan's patch-median residual; the center line is the median and the shaded band spans the interval from the 25th to the 75th percentile within eight fixed equal-width sensor-range bins. No curve is fitted and the band is descriptive rather than a confidence interval. The three curves contain 501, 501, and 484 supported scan-level observations, respectively.

translation error or Patch gain. These results are consistent with a composite residual rather than a single identified shape or timing error.

5.2. Residual RMS is not sufficient, and robust losses are regime dependent

In the controlled exact-reference study, 60 directed cross-family matches satisfy the 5% residual-RMS rule, corresponding to 36 unique unordered pairs. Their median RMS difference is only 0.07 mm, yet the median larger-to-smaller translation-error ratio is 5.96; 77% differ by more than a factor of two and 53% by more than a factor of five. Representative cases are shown in Table 3. Comparable residual RMS therefore does not uniquely determine translation error in these controlled conditions.

Robust response depends on mismatch structure. For D2 and D3, which resemble localized or gross contamination, Huber weighting or trimming often substantially reduces the resulting pose errors. D1 and D5 are coherent perturbations, but their response is heterogeneous: robust weighting can help in some geometry/configuration combinations while substantial pose error remains in others. D4 shows the complementary case in which appreciable residual can coexist with low translation impact. The supported statement is therefore narrower than “robust losses fail on coherent mismatch”: residual-based robustness has no uniform response to structured scan-to-model discrepancy.

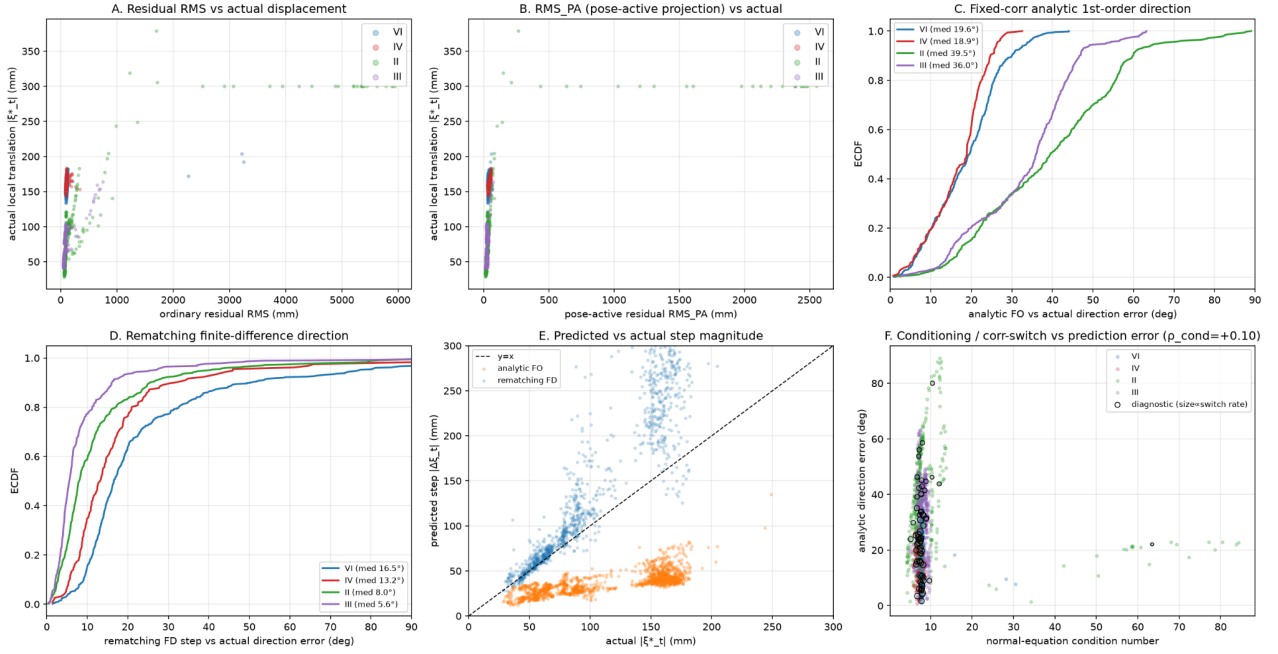


Figure 3: Comparison of local predictions on the same 1456 analyzed real frames. (A) Ordinary residual RMS versus realized local translation displacement. (B) Pose-active residual RMS, RMS_{PA} , versus the same displacement. (C) ECDF of first-order translation-direction error with fixed reference correspondences. (D) ECDF for the finite-difference predictor with rematching. (E) Predicted versus realized translation-step magnitude; the analytic first-order step is systematically smaller than the iterated displacement. (F) Analytic direction error versus normal-equation conditioning; outlined diagnostic points encode correspondence-switch information. VI is development evidence and III remains secondary for this analysis.

Table 3

Representative cross-family conditions matched within 5% residual RMS. RMS and e_i are stored condition medians in millimetres; $|\Delta e_i|$ is the absolute translation-error difference and the ratio is larger-to-smaller e_i . Matching is performed within the same geometry under the frozen 5% rule.

Geometry	Condition A	RMS_A	$e_{i,A}$	Condition B	RMS_B	$e_{i,B}$	$ \Delta e_i $	Ratio
GA	D1 coherent displacement	7.95	8.930	D3 local mismatch	7.70	0.900	8.030	9.9
GB	D4 span growth	4.06	0.058	D3 local mismatch	3.97	0.621	0.563	10.6
GB	D4 span growth	5.33	0.184	D5 coherent tilt	5.08	1.528	1.344	8.3
GC	D3 local mismatch	3.75	0.988	D4 span growth	3.80	0.116	0.872	8.5

5.3. Patch reduces translation error within support, but transfer is not uniform

Patch lowers the marginal median translation error relative to Raw on all four analyzed subsets: 161.4 to 116.8 mm on VI, 160.4 to 126.3 mm on IV, 66.1 to 41.8 mm on II, and 75.0 to 41.3 mm on the 371/1302 retained III frames. Full has lower median translation error than Patch on VI and IV, but higher error on II and III. Thus, the additional structure used by Full does not provide a uniform benefit over Patch. Estimated Full yields translation and rotation errors similar to those of oracle Full when the first registration is initialized at the reference pose; this remains a local comparison rather than a deployment test.

Because the Patch field is calibrated on VI, the primary all-VI result is development evidence rather than an independent validation result. Six-fold LOBO calibration over the six frozen VI view-travel blocks retains essentially the full effect: the out-of-fold field lowers the pooled VI median

translation error from 161.4 to 117.5 mm, with a frame-paired benefit of +43.1 mm (95% block-bootstrap interval [39.6, 44.9] mm), all six held-out blocks positive, and all 501 scans improved. The out-of-fold paired median benefit is approximately 43.1 mm, compared with 43.51 mm in sample, a reduction of about 1%. This reduces concern that the VI development gain is solely an in-sample field-calibration artifact; it does not replace the external evidence from IV/II and secondary III.

The support fractions in Table 2 materially constrain these accuracy numbers. In particular, the 126.3 mm IV Patch median describes only 156/2428 frames. The method should therefore be judged jointly by error and coverage, not by retained-subset accuracy alone.

The II trajectory also limits the claim that spatial detail is always superior to a simpler historical correction. Direct Patch-versus-Global-Vector pairing yields median translation benefits of +33.92, +19.31, +0.24, and +25.03 mm on

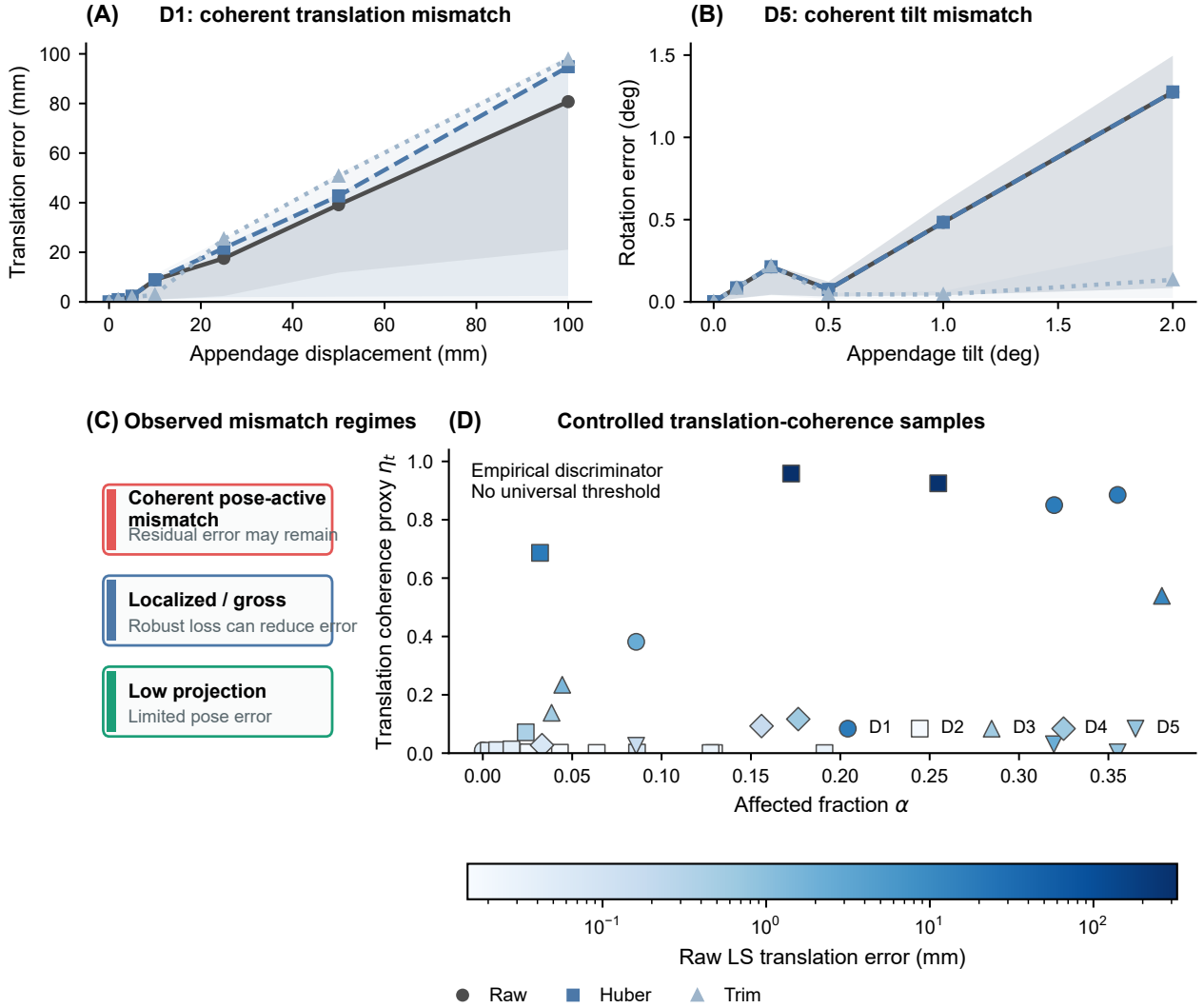


Figure 4: Observed controlled mismatch patterns. (A,B) Point-to-point D1 translation and D5 rotation dose responses. Each curve is the median of the three GA/GB/GC condition medians (20 repetitions per geometry); shading spans those three medians and is not a confidence interval. (C) Qualitative response patterns observed in this sweep; coherent discrepancy can retain a nonzero pose effect. (D) Separately generated phase-map cells summarize 10 repetitions each, plotted by affected fraction and translation coherence and coloured by LS translation error. No universal threshold or exhaustive classification is implied.

VI/IV/II/III; the II whole-block-bootstrap interval is $[-8.49, 18.02]$ mm. Hence II provides no clear evidence that Patch adds a median translation benefit beyond the global historical state vector. A related transfer limitation appears in the view-conditioned state baselines: DBS and Vector-DBS increase translation error relative to Raw on II and III, while Global-Vector retains a smaller positive Raw-relative gain. Full, by contrast, loses its additional benefit relative to Patch on these trajectories. The evidence points to trajectory-dependent transfer of view/spatial specificity rather than a universal advantage from a more detailed historical model.

5.4. Historical transfer and baseline separation

The state-space comparators separate the value of historical supervision from the value of spatial geometry. DBS

predicts a view-conditioned direction and magnitude, Vector-DBS averages historical translation-bias vectors in the same frozen view neighbourhood, and Global-Vector applies one target-wide VI translation-bias vector. DBS and Vector-DBS substantially reduce translation error relative to Raw on VI and IV but increase it on II and III, whereas Global-Vector yields positive paired median reductions on all four analyzed subsets (Table 6). Patch also remains positive relative to Raw on all four. The II result is therefore not evidence that all historical correction fails; it specifically limits the added value of view/spatial specificity under that trajectory.

A fixed 2×2 decomposition separates Full's two implementation changes: view-neighbour conditioning and equal-scan aggregation. With point-count weighting fixed, the median of the framewise translation-error differences between view-conditioned and all-history correction is -40.79 ,

Table 4

VI leave-one-block-out evaluation of the Patch field. The fixed 24-patch partition is retained, but each held-out block is scored with a discrepancy field rebuilt from the other five VI blocks. Benefit is computed within each frame as Raw minus Patch translation error; positive values favor Patch. The pooled interval is a 95% bootstrap interval over the six fixed blocks.

Held-out VI block	<i>n</i>	Raw median (mm)	Patch OOF median (mm)	Paired benefit (mm)	Improved fraction
0 [0,70)	70	151.2	112.0	+40.6	1.00
1 [70,140)	70	151.7	111.0	+38.7	1.00
2 [140,219)	79	165.5	125.0	+41.9	1.00
3 [219,315)	96	168.4	123.6	+45.3	1.00
4 [315,418)	103	164.2	117.9	+44.5	1.00
5 [418,501)	83	159.3	114.7	+44.1	1.00
OOF pooled	501	161.4	117.5	+43.1 [39.6, 44.9]	1.00

Table 5

Point-to-point marginal median reference-pose error with whole-block-bootstrap 95% confidence interval. Each cell reports translation in mm on the first line and rotation in degrees on the second line. VI is development; III is the reserved-trajectory support subset, with method-specific follow-up analyses treated as secondary.

Method	VI	IV	II	III
Raw	161.4 [154.4,166.2] 4.51 [4.09,4.75]	160.4 [157.8,166.3] 4.01 [3.88,4.15]	66.1 [58.0,88.8] 1.77 [1.44,2.08]	75.0 [54.2,85.6] 1.66 [1.33,2.55]
Huber	148.9 [139.2,153.4] 3.95 [3.59,4.30]	132.6 [129.3,138.6] 3.40 [3.24,3.55]	50.9 [36.4,64.3] 1.05 [1.00,1.11]	55.9 [41.2,61.1] 1.05 [0.93,1.38]
Trim	144.8 [130.1,148.9] 3.77 [3.45,4.17]	125.2 [117.6,132.6] 3.22 [2.98,3.29]	47.3 [35.2,57.4] 1.05 [0.96,1.11]	55.0 [43.7,58.0] 1.15 [1.05,1.25]
Global	157.0 [148.1,160.8] 4.51 [4.09,4.75]	153.4 [150.8,159.3] 4.01 [3.88,4.14]	55.2 [46.4,77.3] 1.77 [1.44,2.09]	65.4 [47.6,76.3] 1.66 [1.33,2.55]
Patch	116.8 [112.3,121.3] 4.67 [4.20,5.30]	126.3 [120.5,142.7] 4.00 [3.66,4.38]	41.8 [35.2,70.9] 1.27 [0.79,1.65]	41.3 [24.0,56.6] 2.12 [1.46,2.54]
Full	76.9 [73.6,80.8] 2.88 [2.49,3.24]	82.9 [77.6,111.1] 2.46 [2.22,2.66]	68.9 [58.0,74.8] 2.25 [2.01,2.39]	51.2 [43.9,71.6] 2.64 [2.28,3.15]
Estimated Full	77.1 [74.0,81.5] 2.85 [2.47,3.25]	86.6 [78.7,114.2] 2.27 [2.10,2.52]	70.0 [60.4,76.3] 2.29 [2.11,2.47]	51.4 [43.8,71.2] 2.62 [2.28,3.16]

−38.91, +22.67, and +17.63 mm on VI, IV, II, and III, respectively; negative values favor view conditioning. The same VI/IV-versus-II/III sign reversal remains under equal-scan weighting. Within the view-neighbour branch, changing aggregation alone alters paired median translation error by only +0.18, −0.28, +0.22, and +0.02 mm. Thus, within the implemented factorization, the reversal is associated more strongly with view-neighbour conditioning than with aggregation choice. This is an estimator-level association, not a physical explanation of the transfer failure.

A fixed representative-frame diagnostic further probes why II reverses while IV improves. Representative frames are selected from predefined quantile regions of the frame-wise difference $e_{t,\text{Full}} - e_{t,\text{Patch}}$ (best decile, median neighborhood, and worst decile), not by visual inspection. Among the selected diagnostic frames, II does not show systematically poorer historical-view availability than IV: nearest-view distances span 0.11–0.26 on II versus 0.08–0.26 on IV, patch coverage is 1.00 versus 0.96, kernel effective sample size is approximately 15.7–15.9 on both, and II has no fallback cases in the selected diagnostic frames. The clearest

Table 6

Historical state-space baselines. Each cell reports marginal median translation error (mm) / frame-paired median gain relative to Raw (mm; positive is better). DBS and Vector-DBS use the frozen VI view neighbourhood; Global-Vector uses one target-wide VI translation-bias vector. State-space baselines leave rotation unchanged.

Method	VI	IV	II	III
Raw	161.41 / –	160.42 / –	66.15 / –	75.04 / –
DBS	17.89 / +143.87	19.92 / +139.85	142.97 / −72.05	108.83 / −36.48
Vector-DBS	21.28 / +139.90	21.37 / +138.33	135.67 / −66.42	102.37 / −27.42
Global-Vector	155.94 / +11.19	146.61 / +13.44	51.16 / +17.30	62.35 / +8.42
Patch	116.82 / +43.51	126.26 / +33.63	41.78 / +19.10	41.30 / +28.93

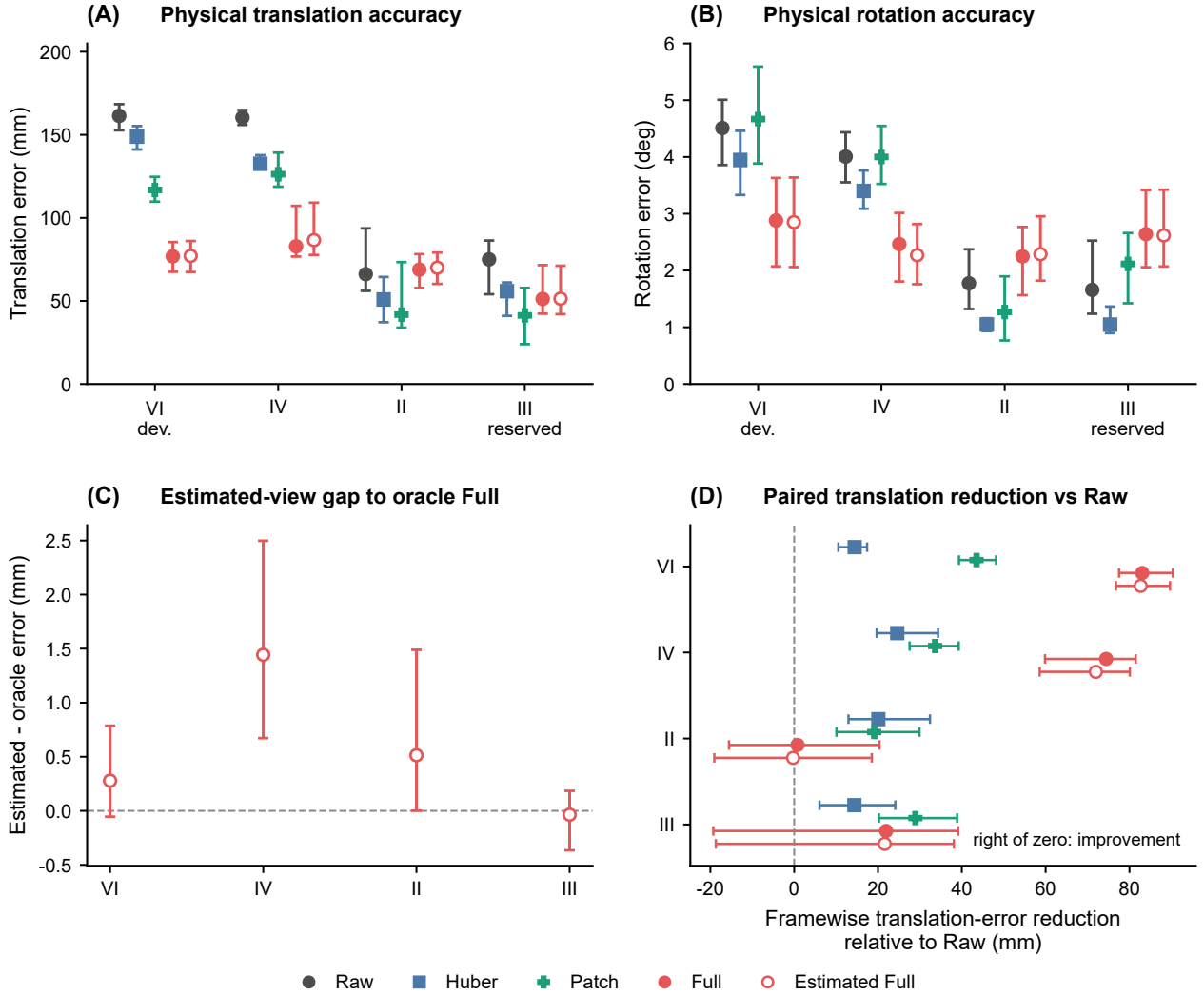


Figure 5: Point-to-point reference-pose error on the common analyzed subsets. (A,B) Marginal medians and descriptive IQRs for Raw, Huber, Patch, Full, and Estimated Full over VI/IV/II/III (501/156/428/371 frames); Trim and Global are reported in Table 5. (C) Median and IQR of frame-paired Estimated-Full-minus-Full translation error; view query and warm start change together. (D) Median and IQR of frame-paired Raw-minus-method translation error; positive values favour correction. Paired medians need not equal differences of marginal medians.

observed difference is directional alignment of the learned correction: the cosine between the predicted and required translation step is $+0.71$ to $+0.85$ across the IV bins, whereas the II median and worst-decile representatives are -0.38 and -0.70 (the II best-decile representative is $+0.76$). In these selected frames, the loss of benefit from view conditioning occurs despite the availability of nearby historical views, but the diagnostics do not uniquely identify its physical cause.

5.5. Robustness, computational cost, and translation-rotation trade-off

The sign of the median translation benefit of Patch over Raw is unchanged across the tested parameter settings. Varying the patch count over $K \in \{12, 18, 24, 32, 48\}$ keeps the frame-paired Raw-minus-Patch translation benefit positive on every trajectory, with ranges of 38.6–56.8 mm

(VI), 31.5–40.6 mm (IV), 14.5–24.3 mm (II), and 23.2–33.5 mm (III). Varying the clustering normal-feature weight over $w \in \{0, 0.15, 0.30, 0.60, 1.00\}$ likewise preserves a positive benefit on every trajectory; pure position-only clustering ($w = 0$) already retains most of the gain. Repeating MiniBatchKMeans with five fixed seeds keeps the benefit strictly positive in all 20 trajectory-seed combinations. Table 7 summarizes these ranges. The primary setting of 24 patches, $w = 0.30$, and seed 42 yields a positive median translation benefit, as do the other tested settings; it is not established as an optimum.

Across the patch-count and normal-weight sweeps, the paired rotation change remains bounded within approximately $\pm 0.66^\circ$, and the five-seed rotation medians vary by at most 0.62° . These checks support stability of the translation conclusion over the tested finite grids; they do not establish cross-target parameter invariance.

Table 7

Sensitivity of the frame-paired Patch-over-Raw translation benefit. Entries are the minimum–maximum frame-paired median benefit (mm) over the stated setting grid; all values are positive. The primary 24-patch / $w = 0.30$ / seed-42 configuration remains frozen and is not re-selected from these sweeps.

Factor	Tested settings	VI	IV	II	III
Patch count K	12, 18, 24, 32, 48	38.6–56.8	31.5–40.6	14.5–24.3	23.2–33.5
Normal weight w	0, 0.15, 0.30, 0.60, 1.00	39.1–57.7	31.8–41.9	17.9–22.2	26.4–32.2
Clustering seed	0, 1, 2, 42, 20240910	43.2–48.1	33.4–36.5	19.1–27.7	26.6–34.8

On the fixed 40-frame diagnostic subsets, the Patch-versus-Raw paired translation benefit remains positive when the iteration cap is increased from 40 to 80 and 160. At 160 iterations, the ordering does not reverse on any trajectory. The initialization study is extended on the frozen 20-frame-per-trajectory roster from the previous 50 mm/2° local regime to 100 mm/5°, 200 mm/10°, and 300 mm/15° using the same four sign-balanced deterministic perturbation directions. The extended evaluation reproduces the results at the previously tested perturbation levels to numerical precision. Patch retains a positive paired translation advantage at every tested level. At 300 mm/15°, the fractions of runs meeting the 50 mm/2° thresholds are 0.20 versus 0.10 on II and 0.33 versus 0.21 on III for Patch and Raw, respectively. On VI and IV, neither method satisfies the thresholds in the tested runs, including those initialized at the reference pose. No numerical failure occurs in the reported 8000 evaluated cells. Both methods begin to encounter the fixed 0.30 m/15° safeguard at the coarsest level, with no material excess clipping for Patch. Patch therefore retains a positive paired median translation benefit under the tested perturbations; on II and III, its fraction of runs meeting the threshold criterion is no lower than Raw's. The criterion is uninformative on VI and IV, and neither method is evaluated as a global acquisition pipeline.

The primary effect directions are also insensitive to the external block-length convention. Re-aggregating the identical retained frames into 25-, 50-, and 100-frame blocks keeps Patch-versus-Raw and Patch+Huber-versus-Huber positive on IV/II/III with 95% block-bootstrap intervals above zero and at least 94% of blocks positive. The II Patch-versus-Global-Vector contrast remains directionally positive but unresolved: its block interval still crosses zero at 50 and 100 frames. No block length is selected to improve significance.

A single-thread CPU benchmark on an Intel Core Ultra 9 285K uses the same frozen 80-frame roster, one warm-up and five timed calls per frame/method, with BLAS operations and nearest-neighbor queries each restricted to one thread. Raw and Patch use static, prebuilt model geometry and KD-trees that are cached outside the per-frame timing window. Under this common accounting boundary, the median per-frame latency is 477.5 ms for Raw and 412.5 ms for Patch (0.86× Raw). The lower Patch median is an empirical benchmark result under the stated accounting boundary; no additional Patch-specific per-frame preprocessing stage is omitted, and this result does not establish

lower asymptotic complexity. Full includes its per-frame view query, corrected-model construction, and KD-tree construction (about 11.8 ms combined) and remains comparable at 427.7 ms (0.90× Raw). Estimated Full includes a nominal registration followed by query/model/tree construction and a second corrected-model registration, yielding 953.3 ms per frame (2.00× Raw). Peak resident memory is 157–160 MB across methods. For Patch, the one-time offline costs are 151.7 ms for 24-patch clustering, 0.18 ms for VI field assembly, 0.91 ms for corrected-model construction, and 11.0 ms for its cached KD-tree. These measurements are hardware-specific cost reports, not a real-time or flight-readiness claim.

Combining spatial correction with robust weighting improves translation further but exposes a six-degree-of-freedom trade-off. Relative to Huber, Patch+Huber gives frame-paired median translation benefits of 45.38, 31.81, 7.87, and 19.66 mm on VI/IV/II/III, while the corresponding medians of the framewise increases in rotation error are 0.421°, 0.052°, 0.498°, and 0.720°. Patch+Trim has lower marginal translation medians than Patch on all four trajectories and has its largest reduction relative to Patch on IV, but its rotation response is likewise trajectory dependent. These combinations are therefore not uniformly superior six-degree-of-freedom estimators.

Two implementation ablations narrow the interpretation of the rotation increase without turning either variant into a new method. Re-estimating model normals on the corrected geometry with the same local-PCA $k = 16$ protocol leaves the point-to-point Patch+Huber result unchanged to numerical precision (maximum per-frame difference 1.4×10^{-14} mm and 8.9×10^{-16} deg), as expected because the point-to-point Kabsch update does not use surface normals. In the point-to-plane diagnostic channel, where normals do enter the system, recomputation reduces the paired rotation penalty relative to Huber from 1.118° to 0.753° on VI, 0.612° to 0.276° on IV, 0.888° to 0.656° on II, and 1.059° to 0.986° on III while preserving the translation gain. This is consistent with stale normals contributing in the point-to-plane formulation only; it does not identify a unique cause. A single neighborhood-smoothing step with k fixed at 16 likewise fails to reduce the point-to-point rotation penalty. Thus the primary point-to-point trade-off cannot be attributed to stale normals, and the tested boundary smoothing does not remove it.

Coarse initialization: initialization level (mm/deg) vs capture and final error

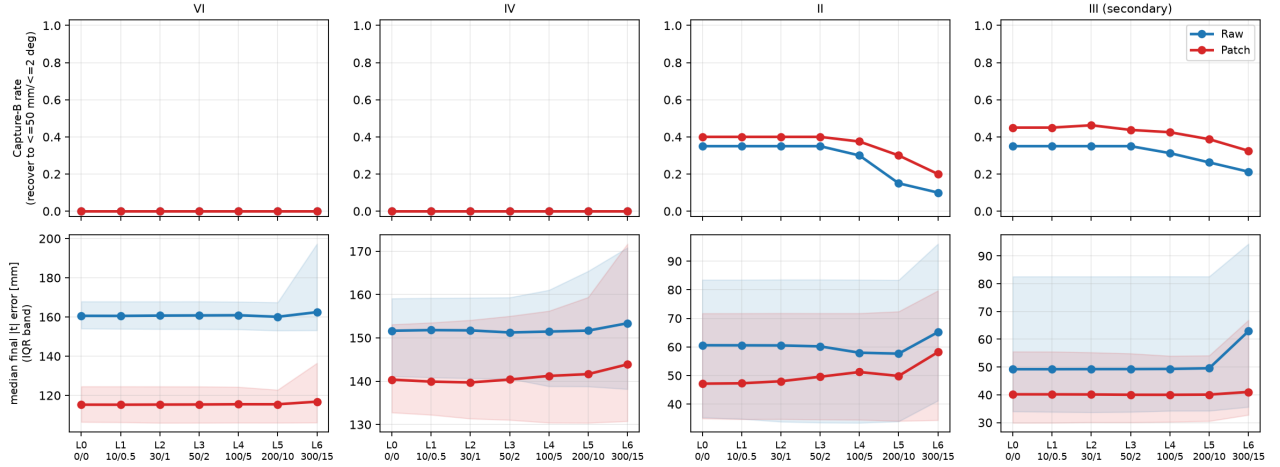


Figure 6: Sensitivity to the tested initialization perturbations on the fixed 20-frame-per-trajectory roster. Initial perturbation increases from the reference pose to 300 mm/15° using the same four sign-balanced deterministic directions at every level. Top: fraction of runs with final errors satisfying $e_t \leq 50$ mm and $e_R \leq 2^\circ$; these are descriptive error thresholds rather than mission tolerances. Bottom: median final translation error with IQR. Across the tested perturbations, Patch retains a lower median translation error than Raw. On II and III, its fraction of runs satisfying the stated error thresholds is no lower than Raw's. On VI and IV, neither method satisfies the thresholds in the tested runs, including those initialized at the reference pose, so the continuous error distributions are more informative for these trajectories.

Table 8

Single-thread computational cost on one CPU workstation over 80 frozen frames (median of five timed calls after warm-up). Raw and Patch use static geometry and KD-trees cached outside the per-frame timing window; Full includes per-frame query/model/KD-tree construction, and Estimated Full includes both registrations. One-time Patch setup costs are reported in the text. Latency and peak RSS are hardware-specific benchmark measurements, not real-time requirements.

Method	Model/tree accounting	Registrations/frame	Median (ms)	p95 (ms)	xRaw	Peak RSS (MB)
Raw	Static nominal model/tree cached; setup excluded per frame	1	477.5	808.1	1.00	158
Patch	Static corrected model/tree cached; setup excluded per frame	1	412.5	611.5	0.86	158
Full	Query-dependent model/tree rebuilt and included per frame	1	427.7	702.0	0.90	158
Estimated Full	Nominal pass plus query-dependent corrected model/tree included	2	953.3	1638.7	2.00	160

6. DISCUSSION AND LIMITATIONS

The main mechanistic result is that scan-to-model residual severity and local pose effect are distinct. The projection in Sec. 3 identifies the residual component in the subspace generated by local rigid motion. The translation direction of the associated first-order pose step agrees more closely with the realized translation direction on VI and IV than on II and III. The finite-difference predictor with rematching gives higher directional agreement, consistent with limitations of the fixed-correspondence approximation. Controlled

matched-RMS cases provide an independent counterexample to treating residual RMS as a sufficient proxy for translation risk.

The historical compensation result should be interpreted more narrowly than a generic “learned bias correction.” Manufacturing scan-to-CAD work shows that rigid registration can absorb geometric deviation, and uncertain-model registration changes how nominal geometry enters the objective. Here, the historical field is calibrated on the same non-cooperative spacecraft target and then applied at geometry level before a new correspondence search. Compared with state-space history methods, Patch retains spatial structure

Table 9

Direct frame-paired translation benefits with whole-block-bootstrap intervals. For each comparison, benefit is computed as baseline error minus target-method error within each frame. Each cell reports the median paired benefit / 95% interval in mm; positive values favor the named target method. VI is development evidence; III is secondary.

Comparison	VI	IV	II	III
Patch vs Global-Vector	+33.92 / [26.69, 43.55]	+19.31 / [12.09, 24.21]	+0.24 / [-8.49, 18.02]	+25.03 / [15.81, 30.36]
Patch+Huber vs Huber	+45.38 / [43.19, 47.06]	+31.81 / [22.40, 34.69]	+7.87 / [1.61, 12.09]	+19.66 / [6.08, 34.16]

Table 10

Marginal median translation / rotation error for Patch and robust combinations (mm / deg). These fixed-parameter combinations are reported to expose the translation–rotation trade-off rather than to select a universally superior estimator.

Method	VI	IV	II	III
Patch	116.8 / 4.67	126.3 / 4.00	41.8 / 1.27	41.3 / 2.12
Huber	148.9 / 3.95	132.6 / 3.40	50.9 / 1.05	55.9 / 1.05
Trim	144.8 / 3.77	125.2 / 3.22	47.3 / 1.05	55.0 / 1.15
Patch+Huber	102.9 / 4.31	102.1 / 3.54	40.1 / 1.53	32.2 / 1.73
Patch+Trim	96.1 / 4.48	89.9 / 3.45	40.5 / 1.63	37.6 / 2.16

and can change both translation and rotation through re-matching. This is the practical distinction from post-hoc pose correction and the reason the translation–rotation trade-off must be reported explicitly.

Patch does not establish universal superiority over simpler history. On II, Patch reduces translation error relative to Raw, but its paired median benefit over Global-Vector is only 0.24 mm, with a confidence interval spanning zero. Separately, the representative-frame analysis of Full versus Patch shows that the loss of benefit from view conditioning can occur despite nearby historical views. In the selected diagnostic frames, the learned view-conditioned correction is positively aligned with the required step on IV but becomes misaligned in the median and worst II bins. This is an estimator-level association and does not uniquely identify the physical origin of the mismatch or establish a universal failure mode.

The Patch+Huber rotation penalty matters for operations that require joint pose accuracy. For scale, a rotation of 0.720° (≈ 0.0126 rad) produces a displacement of approximately 12.6 mm at a 1 m lever arm perpendicular to the rotation axis, or 25.1 mm at 2 m. This illustrates the geometric scale of the reported angular penalty; it does not measure the change in end-effector position error. Whether that is acceptable depends on the task: inspection can tolerate different orientation errors from docking, contact-rich capture, or robotic grasping. Because no mission-specific tolerance was prespecified, the paper does not classify the observed rotation costs as acceptable or unacceptable; it reports them as an engineering trade-off requiring task-level validation.

The rotation-cost mechanism is narrowed but not uniquely identified. Stale model normals cannot explain the primary point-to-point rotation penalty: the solver does not use normals, and recomputing them leaves the outputs unchanged to numerical precision. The tested one-step boundary smoothing also does not reduce the point-to-point

penalty. In contrast, normal recomputation partially reduces the point-to-plane penalty, which is consistent with stale normals contributing in that diagnostic formulation. In the primary pipeline, correspondence rematching, displacement of the target geometry, and robust reweighting remain coupled factors and are not separately identifiable from the present ablations. The remaining point-to-point rotation cost is therefore reported as an empirical trade-off rather than assigned to a single mechanism.

6.1. Validation and deployment scope

The validation scope is deliberately narrower than a deployment study. The main accuracy experiments use one point-to-point ICP back end; point-to-plane is used only as a mechanism diagnostic, and no claim is made of superiority over generalized, certifiable, or global registration pipelines. The Patch field is target-specific and the primary field is fitted from all 501 VI calibration scans, so VI remains development evidence; however, six-fold VI LOBO retains a +43.1 mm paired benefit with all six held-out blocks positive, reducing concern that the development result is solely an in-sample calibration artifact. The external IV/II/III trajectories provide cross-trajectory transfer evidence only within the fixed retained support; they do not establish generalization to other targets, target configurations, calibrations, or LiDAR sensors. The computational cost of rebuilding the frozen partition/field is small on the tested workstation, but whether a rebuilt field remains valid after an actual target, configuration, or calibration change is not evaluated.

The sensitivity analysis covers only the stated grids for patch count, normal-feature weight, clustering seed, iteration budget, initialization perturbation, and block length; it does not sweep the Huber constant, MAD floor, Trim fraction, support radius, fallback logic, distance gating, sampling policy, clustering family, or every view-kernel setting. These

tests establish sign consistency of the Patch-over-Raw translation benefit within the evaluated settings; the magnitude of the benefit, rotation cost, and failure behavior may differ outside them. Likewise, no exhaustive frame-level taxonomy of boundary stops, fallback events, outliers, or other failure states is reported across every sensitivity arm.

The support rule is computed from the dataset reference view and serves as an experimental inclusion rule, not an online confidence detector. A deployable system would require a support/novelty criterion computable from available onboard quantities. Computational cost is benchmarked transparently on one single-thread CPU workstation. Raw and Patch share the same timing boundary, with static geometry and KD-trees cached outside the per-frame window; the lower Patch median is an empirical result under the shared timing boundary rather than a consequence of omitted per-frame Patch preprocessing. Full includes its query-dependent model/tree construction and remains comparable to Raw, whereas Estimated Full is approximately twice as slow because it performs two registrations. These hardware-specific measurements do not establish real-time or flight-ready performance. The block-length sensitivity analysis changes the aggregation convention but does not explicitly model within-trajectory temporal autocorrelation or establish block independence.

Initialization evidence remains bounded even after the coarse extension. The fixed perturbation study now reaches 300 mm/15°. Patch retains a positive paired median translation benefit under the tested perturbations. On II and III, its fraction of runs meeting the 50 mm/2° criterion is no lower than Raw's; the criterion is uninformative on VI and IV. Both methods begin to meet the fixed 0.30 m/15° safeguard at the coarsest level. Estimated Full still substitutes an estimated view only after a reference-initialized first pass. No global initializer, arbitrary non-reference acquisition study beyond the fixed safeguard, or end-to-end front-end pipeline is evaluated, so global acquisition remains unevaluated.

Coverage and statistics impose additional limits. Only 6.4%, 34.2%, and 28.5% of IV/II/III satisfy the fixed support rule, so retained-subset accuracy cannot be generalized to excluded frames or complete trajectories. The 6/4/9/8 primary analysis blocks also provide limited power; sign tests and bootstrap intervals remain conditional summaries rather than population-level guarantees. Re-aggregation with 25-, 50-, and 100-frame blocks preserves the direction and above-zero block-bootstrap intervals of the Patch-versus-Raw and Patch+Huber-versus-Huber contrasts on IV/II/III, reducing sensitivity to the particular 50-frame convention. It does not resolve the II Patch-versus-Global-Vector contrast, whose interval still crosses zero at the larger block lengths. The block-length check does not estimate the trajectory autocorrelation structure or prove block independence, and no multiplicity-adjusted confirmatory claim is made.

Finally, the EPOS residual source is not uniquely identified. Frame-level reference poses are available, but verified per-point timestamps needed to reconstruct intra-scan motion are unavailable for this analysis. Reference-pose

uncertainty, extrinsic calibration, scan integration, surface response, and sampling can therefore contribute alongside model discrepancy. The controlled exact-reference experiments demonstrate that the implemented structured mismatches are sufficient to create local pose effects; they do not prove that the same physical mechanism dominates EPOS-Lid.

7. CONCLUSIONS

Structured scan-to-model discrepancy can induce local pose changes that residual RMS alone does not characterize. On the analyzed EPOS-Lid frames, the first-order step with fixed correspondences approximates the realized translation direction, while the finite-difference predictor with rematching provides closer directional agreement. A spatial discrepancy field calibrated on trajectory VI reduces median reference-relative translation error compared with uncorrected ICP on all four main analysis subsets. This benefit persists under leave-one-block-out field calibration on VI and across the tested parameter and initialization settings.

The findings remain conditional on the calibrated target and retained support, which covers 6.4%, 34.2%, and 28.5% of IV, II, and III, respectively. Patch provides no clear additional median translation benefit over Global-Vector on II, and view conditioning does not improve performance uniformly. Combining Patch with robust weighting can further reduce translation error while increasing rotation error. The study therefore supports local translation compensation under the evaluated conditions, without establishing full-trajectory coverage, cross-target generalization, global acquisition, or uniformly improved six-degree-of-freedom accuracy.

DATA AND CODE AVAILABILITY

The public EPOS-Lid dataset is described by Renaut et al. [4] and remains subject to its original access and licensing terms. The study-specific implementation, fixed frame/block manifests, patch assignments and field, per-frame outputs, statistical scripts, and figure/table generation scripts are not included in the third-party dataset. The study-generated results are retained in an internal archive with framewise outputs, exact commands, configurations, random seeds, and SHA-256 hashes of their input and output assets. These materials are not publicly archived and are available from the corresponding authors upon reasonable request; independent reproduction is therefore limited until a public or reviewer-accessible archive is released.

CRedit AUTHORSHIP CONTRIBUTION STATEMENT

Jiayao Li: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. Xinglin Ren: Methodology, Validation, Formal analysis, Writing – review & editing.

Jipeng Huang: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing. Haichao Sun: Resources, Supervision, Writing – review & editing.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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